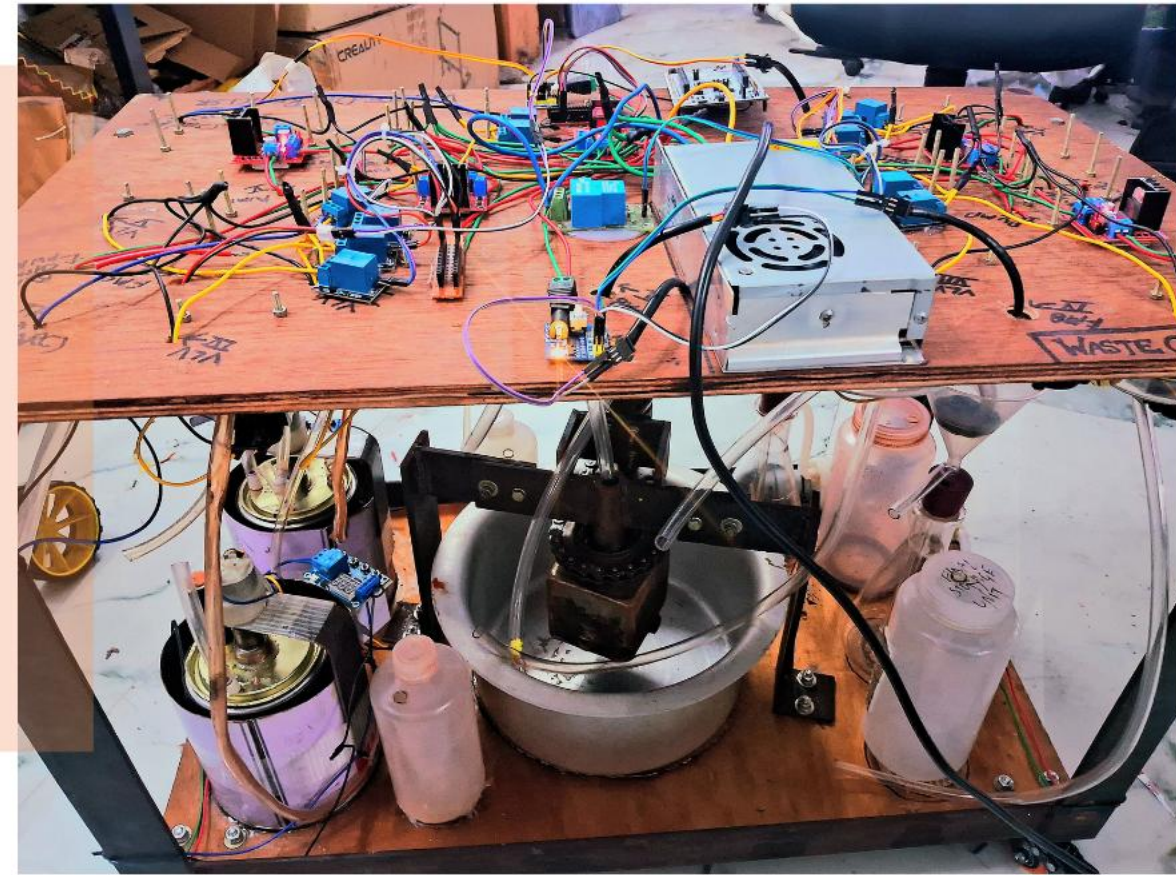


Team **Oillovators** presents....

WASTE OIL RECYCLER

*"Revolutionizing Oil Recycling -
Because Every Drop Matters!"*



College: Institute Of Engineering & Management, Kolkata

Team Members: Shyantam Kundu

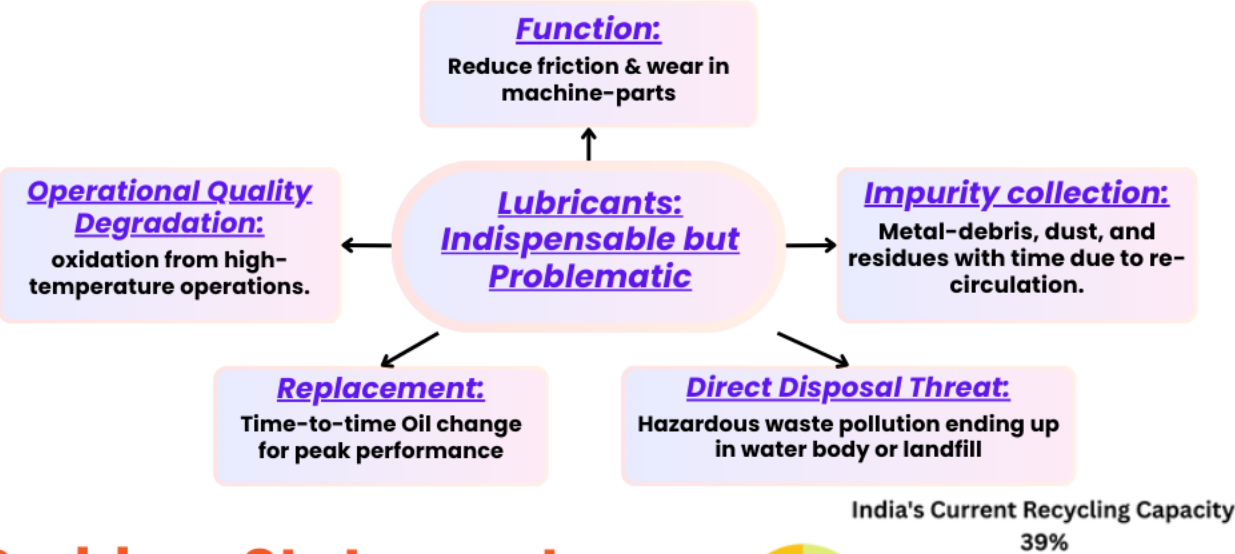
Team Lead Name: Shyantam Kundu

Business Type: Product based service

| **Team Lead Contact Info:** shyantan5@gmail.com, 9123954154

| **Industry:** Waste Management, Alternative Fuel

Problem Statement: The Vexed Crisis of Used Lubricants



Problem Statement

- ☛ Globally, **5 liters of waste engine oil** is generated every six months by one car, worsening environmental crises.
- ☛ In India, **4.3 billion liters** of used lubricant are produced annually, with only **39% recycled** by 223 out of 257 facilities.
- ☛ With automotive sector sales rising **20% YoY** and lubricant sales projected to grow at **4.76% CAGR** to **3.15 billion liters by 2026**.
- ☛ So, effective waste management of Used Oil is necessarily implemented for enforcing Environmental Protection.
- ☛ This will further help Lubricant industry realize their **EPR and CSR objectives** crucial for sustainability.

Key Statistic Point	Scale of the Problem
Oil usage per vehicle	Cars: 4–7.6L , Bikes: 0.5–2L , Trucks/Buses: 10–20L
India's vehicle count	7.87 Cr 4-wheelers, 24.73 Cr 2-wheelers, 4.83 Cr commercial vehicles
High Displacement Engines	4,543 diesel locomotives; Generators & Marine engines need 1,000L to 15,000L+ Oil / change.
Industrial sector	36,000 industries produce 2.048 Billion Litres of Used Lubricants (ULO) annually
Transportation sector	All vehicles in total produce 2.24 billion Litres of Used Engine Oil (UEO) annually.
Total ULO and recycling	India generates 4.3 billion Ltrs of ULO + UEO annually & only 1.39 MMT / 39% [1] is recycled by 223 Used Oil recycling facilities [4] out of 257 Spent Oil recycling Facilities.

TOPIC OF THE PLAN SOLUTION

From the problems presented it is evident that:

A Huge Gap in India's Oil Recycling Sector to Bridge



Reaching to Minutest Levels of Used Oil garages of Interior India



Effective Recycling and Re-purposing of the Recycled Product



Hence, introducing our solution “Waste Oil Recycler”

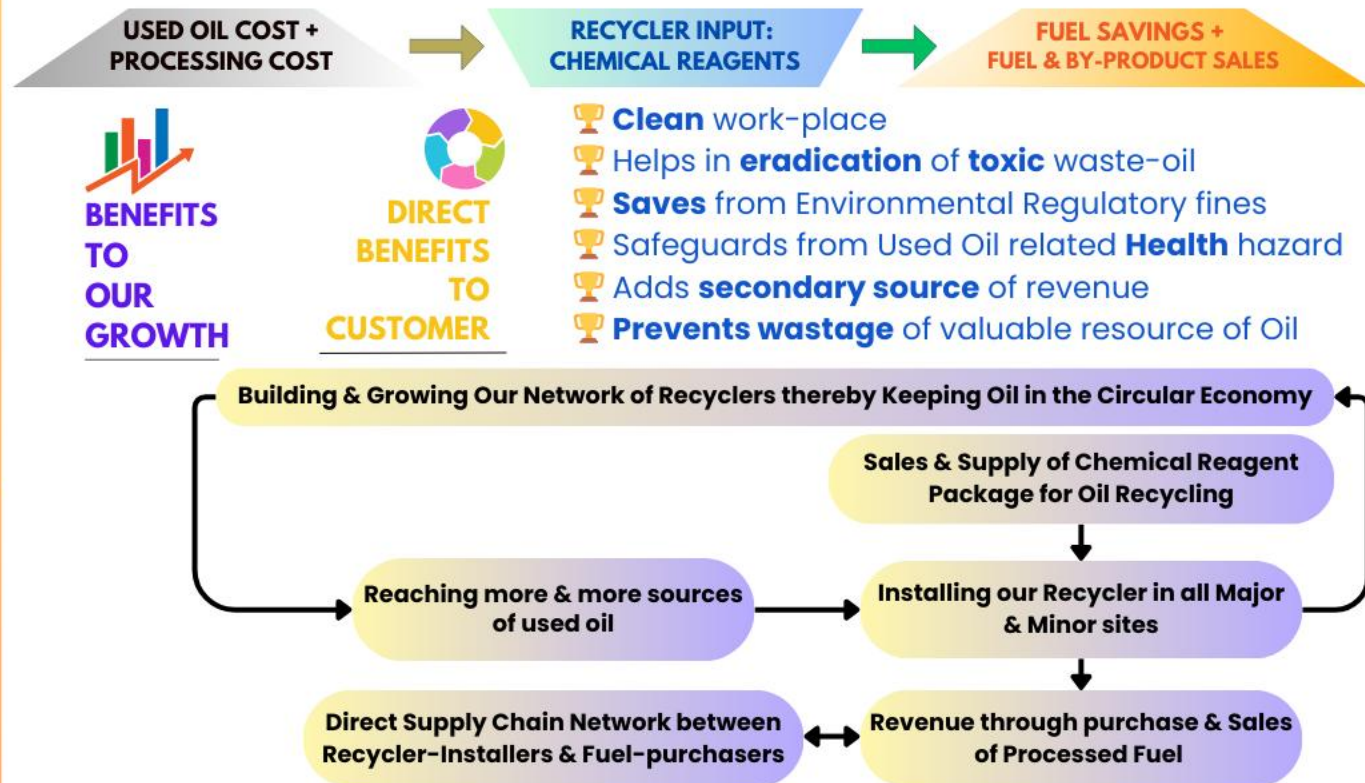
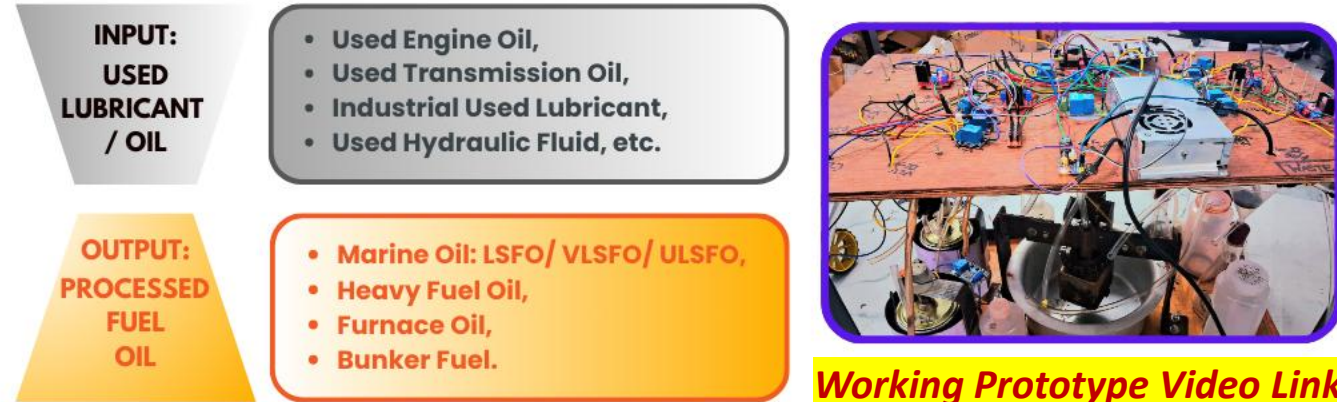
By-products generated 20%



Treated Fuel Oil 80%

Features of the Solution:

- ✓ **Diverse** Used Lubricant Oil **Input**.
- ✓ **Cost-efficient** recycling **mechanism**.
- ✓ **Specially** Curated **Non-toxic** Recycling **process**.
- ✓ **Maximized** Recycling **Output** of Fuel-Oil.
- ✓ **Re-usable** Fuel Oil to meet **Industrial** Heavy **Fuel demands**.
- ✓ **Minimal By-product** produced also re-usable.
- ✓ **Modular** design & Full **Autonomous Operation**.
- ✓ **Reasonable** Up-front **Cost**, **High** profit **Returns**.
- ✓ **Less** Maintenance, **Durable** Parts & **Low** Operating **Cost**.



8 Stages of Recycling

1. Collection of Used Engine Oil
2. Initial Filtration
3. Oil Acidification
4. Alcoholic Base Preparation
5. Trans-esterification
: acidic oil+ alcoholic base
6. Centrifugation
7. Final filtration
8. Extraction & Final storage



MASS COLLECTION section

Used Lubricant-oil collected from various sources is stored here.



FILTRATION section

Oil is filtered twice for purity to separate larger impurities.



FINAL STORAGE section

Coolant material cools engine oil via a copper pipe.



CENTRIFUGE system

High RPM centrifugation for quick decantation.



BOILER + BLENDER section



Used-Oil treated with cleansing-acid & Alcohol at a set temperature.

CHEMICAL storage



Required Chemicals are stored in HDPE containers.

Working Prototype
Video Link



[BEFORE]
Used Engine Oil
from Bike



[AFTER]
Fuel Oil treated
from WOR



Customers and Use Cases

USE CASE 1  Installers Being Potential Users: Self utilization of fuel oil after recycling reduces cost of Fuel purchase.

-  Mining Sites – Trucks :engine oil; heavy machinery runs on HFO.
-  Construction Sites– Vehicles: engine oil;large equipment : HFO.
-  Railway Depots – Locomotives use engine oil; others use HFO.
-  Ports & Shipyards – Trucks need engine oil; ships run on HFO.
-  Industrial Zones – Factories use HFO; logistics-fleet need engine oil.



USE CASE 2  Installers Being Potential Producers: Selling recycled fuel oil to earn revenue.

-  Used by local garages and service centers to reuse UEO.
-  Truck depots and bus terminals.
-  Agricultural farms to reuse UEO from tractors and harvesters.



POTENTIAL SITES OF RECYCLER INSTALLATION

- Collaborating with OEM Automotive Service Centres: 2/3/4/Multi-wheeler
- Local Garages, Local Oil Recyclers & Machining Industries
- Railway yards & Shipyards , Bus & Truck Depots
- Construction & Mining Sites

POTENTIAL USERS OF PROCESSED FUEL-OIL

- Marine Oil: LSFO/ VLSFO/ ULSFO: Ships, Water-Vessels
- Heavy Fuel Oil: Industrial Engines, Diesel Generators, Fuel for Heavy Equipment machinery
- Furnace Oil: Industrial Furnace
- Bunker Fuel: Industrial Boilers



VALUE PROPOSITION CANVAS

VALUE PROPOSITION



CUSTOMER SEGMENT



Waste Oil Recycler

- Lower operational costs**
- Reduces disposal and recycling costs.
 - Enhances profitability from waste oil.

KEY DIFFERENTIATORS & VALUE CREATION

- Modular, scalable design:**
- Flexible for any business size.
 - Grows with operational needs.

- Regulatory compliance**
- Easy adherence to waste laws.
 - Avoids penalties for violations.

- Turn-Key Solution:**
- No need for specialized Supervision.
 - Simple setup, minimal maintenance.

- New revenue streams**
- Generates fuel from waste oil.
 - By-products for road construction.

OVERVIEW OF MARKETING PLAN :
BUSINESS MODEL



Business Model Canvas

Idea/Innovation Title: Waste Oil Recycler

Key Partners • OEM Service centers • Private Garages • Govt. Railway yards • Local oil recyclers • Small-scale industries • Medium-scale industries • Heavy industries • Used Oil collectors & sellers • Construction & Mining sites • Potential sources of lubricant waste 7	Key Activities • Research and development for product improvement and innovation • Manufacturing and production of WOR machines • Marketing and awareness campaigns in target regions • Partnering with local entities for distribution and support 8	Value Proposition • Converts used lubricant/oil into re-usable fuel oil • Optimizes resource utilization and minimizes waste production • Reduces environmental pollution and carbon footprint • Provides a cost-effective and Eco-friendly solution • Enhances sustainability in the Auto sector and industries 2	Customer Relationships • Provide educational content on the benefits and usage of the WOR machine • Offer customer support for installation, maintenance, and troubleshooting. • Gather feedback and insights • Build relationships through service contracts and ongoing support. 4	Customer Segments • Marine Oil sector: LSFO/ VLSFO/ VLSFO • Industrial fuel needs: Heavy Fuel Oil (HFO) • DG Set Operators • Bunker Fuel: Boilers • Furnace Oil: Furnace • All Key partners, especially: • Garages & Service centres • Shipyards & Railway yards • Construction & Mining sites • Small to Heavy scale industries 1
Key Resources • Technological expertise in recycling and biofuel production • Manufacturing facilities and supply chain management • Collaborative partnerships with NGOs, retailers, and local distributors • Marketing and sales teams for promotion and distribution 6			Channels • Direct sales to end-users through online and offline channels • Partnerships with NGOs and local distributors • Sales through government and industry contacts • Online platforms for awareness, education, and direct sales 3	

Cost Structure

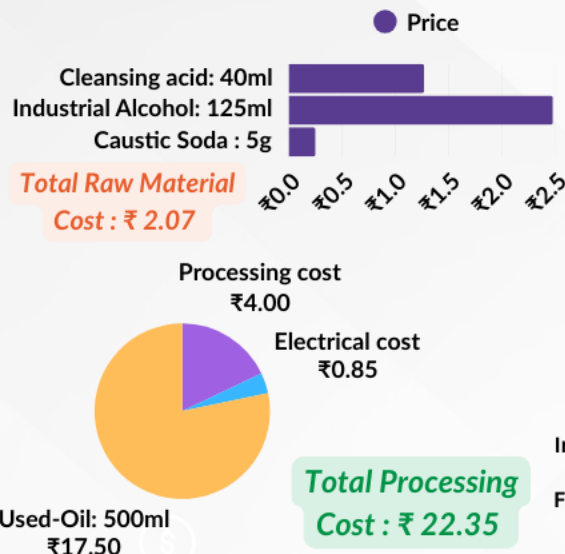
• Research and development expenses for product enhancement • Manufacturing costs for fabricating the WOR machines • Marketing and distribution expenses for reaching target customers • Customer support and maintenance costs • Operational costs for maintaining production facilities 9
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Revenue Streams

• Direct sales of WOR machines • Licensing or partnership agreements for wider distribution • Sales of processed biofuel from secondary revenue streams • Potential subscription-based models for ongoing maintenance and support 5
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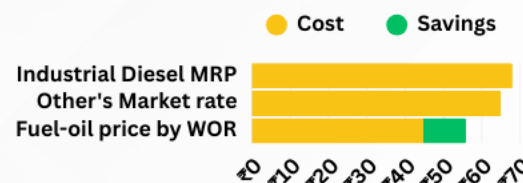
COST ANALYSIS

Current Prototype Specifications	
Processing Capacity	500mL
Time taken to process 1 batch	45 mins
Power Rating	180 watts



PROFIT ANALYSIS

Processing Cost : 500mL	₹ 22.35
Selling Price : 500mL (@ ₹60/L)	₹ 30
Profit	34.23%
Less than Market rate by (@ ₹65/L)	7.69%



COMPETITION ANALYSIS

Aspect	ZYG Plant	KLEEN SMFS	JENSEN	Waste Oil Recycler
Temperature Hazard	300-380°C	High temp.	N/A	100°C
Air pollution	Flue gas	None	N/A	None
No. of operators	2 workers	Portable	Single	Single
Electrical consumption	High	Efficient	Expensive	Low
Size & portability	Large	Large	Portable	Small & Portable
Total Cost of Ownership	High	High	High	Low
Pre-heat	N/A	Preset	N/A	None

RECYCLER Capacity	RECYCLER COST	Volume of OIL to be recycled	Processing Cost	Selling Cost (@ ₹60/L)	Profit	No. of Cars (each car drains avg. 4-6L used-oil)	No. of Days needed (for a repair facility handling 10-20 cars/day)	Time required for recovering Investment
500ml/45mins	₹ 25000	1650L	₹ 73755	₹ 99000	₹ 25245	330	98 days	3 months 2 weeks
5L/45mins*	₹ 70000	4640L	₹ 207408	₹ 278400	₹ 70992	928	42 days	1 month 2 weeks

INNOVATION STACK

Working Prototype Video Link



Thermo-Chemical Stirring

Efficiently mixes chemicals during recycling with automated stirring and precise thermal control.



Advanced Catalytic conversion

Uses special catalyst blend to speed up chemical reactions, improving contaminant breakdown and boosting recycled oil purity.



Rotary Phase Separator

Reduces decantation time, rapidly separates oil from sludge.



Marketing Plan: Sales Channels & Key Market Segments

OIL REFINING SERVICES



REVENUE



\$STREAM\$



RECYCLER SALES

PRIMARY POINTS OF SALES:

- Waste-oil refining centres:
Small scale & Local
- Authorised Service Centres:
 >> Automotive repair Centre
 >> Fleet maintenance Centers

SECONDARY POINTS OF SALES:

- Local automotive repair shops:
2/3/4-wheelers, Bus & Trucks
- Other vehicles repair Centre:
Locomotives & Water-vessels
- Industrial machines:
Machining, Steel industry
Construction & Mining Sites

CONSUMABLES SALES & UPKEEP FEES

CHEMICAL REAGENT PACKAGE SALES:

- Reasonably priced package
- Inclusive of all Chemical Reagents
- On-time chargeable delivery

RECYCLER MAINTENANCE:

- Maintenance Visit Charge
- Replaceable Parts Cost
- Damaged Part Repair-cost

Consumables SALES

Upkeep Fees

Consumables SALES

Upkeep Fees

Refining Services

RECYCLER SALES

RECYCLER SALES

Brand Reinforcement
10%

Partnerships
20%

Network Expansion
40%

Service Offerings Enhancement
30%

Targeting OEM Service Centres:

- Car: 131,295 [6]
- Bike: 54,267 [7] &
- Truck and bus 3,271 [8].

Potential revenue from Increased Lubricant sales: **Up to ₹80 crores.**

- Increased Customer Loyalty
- Good relations with Sellers of Respective Lubricant Brands
- New Market opportunities
- Good Public Reach

Solidify THE OIL Company as a brand caring towards Nature and meeting its CSR & EPR : **Up to ₹20 crores.**

Forming partnerships with OEMs and independent Service centers via:

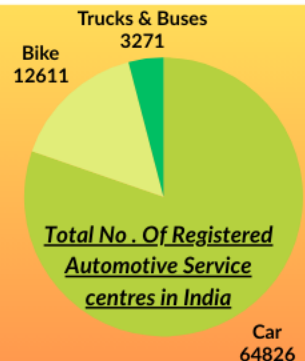
- Licensing fees,
- Sales Commission.

Potential revenue from partnerships: **Up to ₹40 crores.**

Selling of Recyclers as a value addition for:-

- Service centers,
- Bus depots, Truck Garages
- Local Oil Recyclers, Small-scale Industries
- Any OTHER potential Lubricant Waste source

Sales of Processed fuel oil from secondary revenue streams: **Up to ₹60 crores.**



Financials :Market Size Estimation & Financial Projections

TOTAL ADDRESSABLE MARKET: INDIA

TOTAL NO. OF AUTOMOTIVE SERVICE CENTRES IN INDIA:

188833

TOTAL NO. OF INDUSTRIES IN INDIA: 36185

INDIA'S WASTE OIL MARKET(2023): ₹19,920 CR
GROWTH RATE OF 8.53% CAGR (2025 TO 2030).

SERVICE ADDRESSABLE MARKET: WB

WEST BENGAL'S CONTRIBUTION (~6.3% OF INDIA): ₹1251 CR.

TARGET SECTORS (~60% OF WASTE OIL GENERATION): ₹750 CR.

- AUTOMOTIVE SERVICE CENTERS/GARAGES: ~7,004.
- RAILWAYS/SHIPYARDS: ~50 FACILITIES (EASTERN RAILWAYS HUB, KOLKATA).
- MINING/CONSTRUCTION SITES: ~200 ACTIVE SITES.
- MANUFACTURING INDUSTRIES: ~1,000 UNITS (STEEL, TEXTILES, ETC.).
- MARINE ENGINES & DG SETS: ~500 UNITS.

SERVICE OBTAINABLE MARKET

TOTAL SOM:
(~2.25% SAM):
₹16.9CR

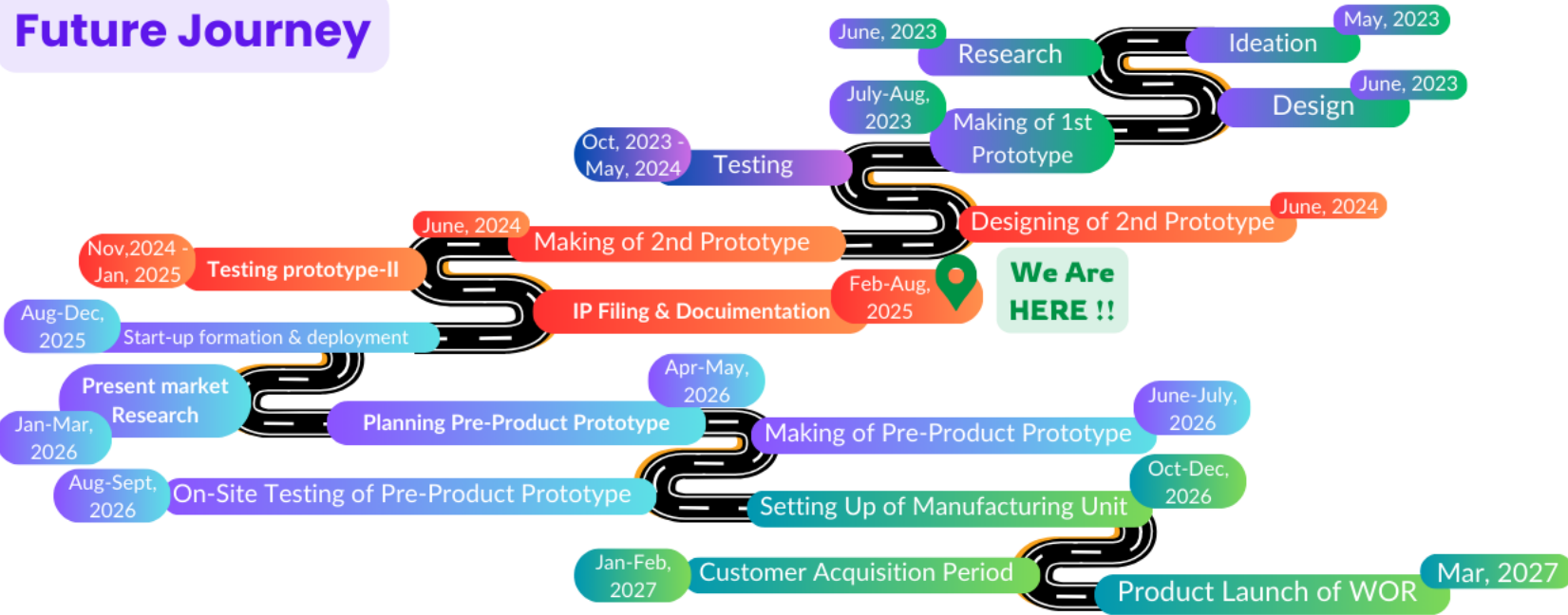
INITIAL MARKET
CAPTURE:
(~0.2% SAM):
₹1.54CR

Category	Details / Remarks	Figures
Initial Investment	Total (CapEx + OpEx)	₹1.58 Cr
CapEx (R&D + Plant Setup)	₹10 L + ₹60 L	₹ 70 L
OpEx (Raw Materials + Labor + Utilities + Marketing + Miscellaneous)	₹33.25 L + ₹20 L + ₹10 L + ₹15 L + ₹5 L	₹ 83.25 L
Production Capacity		
1L Machines: 250 units	Cost: ₹ 33,250/unit	Selling Price: ₹ 40,000 /unit
5L Machines: 150 units	Cost: ₹ 57,500/unit	Selling Price: ₹ 67,500 /unit
10L Machines: 70 units	Cost: ₹ 85,000/unit	Selling Price: ₹ 1,08,000 /unit
Revenue Streams (Year 1)	WOR Sales Revenue (63% Sales Realised) + Consumables Revenue + Service Contracts	(157*40k+94*67.5k+44*108k)= ₹1.75 Cr ₹1.75 Cr + ₹10 L + ₹15 L = ₹2 Cr
Total Revenue		₹2 Cr
Cost of Goods Sold		₹52.5 L + ₹54 L + ₹37.5 L = ₹1.44 Cr
Gross Profit	Total Revenue – COGS	₹2 Cr - ₹1.44 Cr = ₹56 L
Net Profit/Loss (after ₹7 L depreciation)		₹56 L - ₹83.25 L - ₹7 L = -₹34.25 L
Net Profit Margin		- ₹0.48 Cr / ₹ 1.99 Cr = -24.1%
Break-Even Point	Total Investment / Annual Net Profit	~2.5 years
Key Highlights	Affordable machine pricing ensures market entry	1L: ₹40K, 5L: ₹70K, 10L: ₹1.08L
Revenue boosted via consumables & service sales -		Consumables (₹10 L), Service Contracts (₹15 L) contribute 12.5% of revenue

Year	Targeted SOM	Projected Revenue	WOR Units Sold	Consumables + Service Revenue	Total Revenue	Growth Rate	Net Profit
1	0.21% of ₹750 Cr	₹1.54 Cr	295 WOR Units: ₹ 1.75 Cr	₹10 L + ₹15 L	₹2 Cr	-	-₹34.25 L
2	0.25% of ₹750 Cr	₹2.3 Cr	500 WOR Units: ₹ 2.96 Cr	₹39 L	₹3.35 Cr	~67.5%	₹20 L
3	0.8% of ₹750 Cr	₹6 Cr	800 WOR Units: ₹ 5.8 Cr	₹96 L	₹6.76 Cr	~100%	₹1.28 Cr

Current stage: Traction & Progress

Future Journey



NABL Lab Test Report of Fuel Oil Recycled from WOR

Advanced Research & Testing Laboratory
ACCREDITED BY NABL (ISO/IEC 17025:2017)
AN ISO 9001 : 2015 Certified Company
(GOVT. REGISTERED)

Certificate No. - GACB5131
Govt. Registration No. - 78825

TEST REPORT
SAMPLE NOT DRAWN BY US

Test Report no: ARTL/24-25/O-S/048
Date of report issue: 17-06-2024

Customer Name: SHYANTAN KUNDU
Customer Address: 30, MAHADEV APARTMENT, FLAT 4A, NANDANPARA ROAD, NEAR MILAN SAMITI CLUB, MANICKPUR, AIRPORT 2 NO. GATE, DUMDUM, KOLKATA - 700079

Description of Sample: LIGHT DIESEL OIL (LDO)
Date of sample Receipt : 11-06-2024

Sample Mark: NIL
Analysis Started on : 11-06-2024

Status of Sampling: Sample was submitted by the Client.
Analysis completed on : 17-06-2024

SL.NO.	Test Parameters	Test Method	Unit	Result	Limit as per IS 15770: 2008: RA. 2019
01.	Total Acidity	IS: 1448 (Part - 2)	mgKOH/g	2.64	NIL
02.	Ash percent by mass, Max	IS: 1448 - Part - 4	%	0.42	0.02
03.	Carbon Residue: percent by mass, Max	IS: 1448 - Part - 8	%	0.12	1.5
04.	Total Sulphur: percent by mass, Max	IS: 1448 - Part - 33	%	0.0024	1.5
05.	Water Content: percent by volume, Max	IS: 1448 - Part - 40	%	<0.01	0.25
06.	Density @15°C	IS: 1448 (Part - 16)	Kg/m3	0.9894368	---
07.	Flash Point °C, Min	IS: 1448 (Part - 21)	°C	230	66
08.	Pour Point °C, Max	IS: 1448 (Part - 10)	°C	-6	a) Winter (12°C) b) Summer (21°C)
09.	Cloud Point	---	°C	---	---
10.	K. Viscosity @40°C	---	---	18.9464	---
11.	K. Viscosity @100°C	ASTM D445	Cst	10.9629	---
12.	Iron as Fe	AAS Method	%	0.021	---
13.	Copper as Cu	AAS Method	%	0.008	---
14.	Viscosity	IS: 1448 (Part - 25)	Cst	10.9629	---
15.	Sediment percent by mass, Max	IS: 1448 (Part - 30)	%	0.0049	0.10
16.	Calorific Value	Bomb Calorimeter	Kcal/kg	10930	---
17.	Residue on Ignition (Sulphated Ash)	---	%	0.007	---

*****End of Test Report*****

Quality Manager
Authorized Signatory

The test report shall not be reproduced, except in full, without written approval of the company.
Results related only to the parameters tested.

LABORATORY ADDRESS: 12, MONDAL PARA 1ST LANE (2ND FLOOR), KOLKATA - 700 031
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Barunav Kundu

Thank You